

The Wronskian–Pandrosion Calculus

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Abstract

We develop a Wronskian calculus for the Pandrosion quotients $Q(\alpha_j, z) = P(z)/(z - \alpha_j)$. The fundamental identity is $W(P, Q_j) = Q_j^2$; summing over roots gives $W(P, P') = -\sum Q_j^2$, recovering the Turán–Pandrosion identity of Paper 56. For cross terms, $W(Q_i, Q_j) = (\alpha_i - \alpha_j)R_{ij}^2$ where $R_{ij} = P/((z - \alpha_i)(z - \alpha_j))$ is the double quotient. The antisymmetric matrix $\mathbf{W}_{ij}$ encodes the full interaction structure of the Pandrosion fiber, and its Pfaffian relates to the discriminant.

1 The fundamental Wronskian identity

Theorem 1.1 (Wronskian of P with its quotient). *For $P(z) = (z - \alpha)Q(\alpha, z)$:*

$$\boxed{W(P, Q(\alpha, z)) = Q(\alpha, z)^2}, \quad (1)$$

where $W(f, g) = f'g - fg'$.

Proof. Since $P = (z - \alpha)Q$: $P' = Q + (z - \alpha)Q'$. Therefore:

$$\begin{aligned} W(P, Q) &= P' \cdot Q - P \cdot Q' \\ &= (Q + (z - \alpha)Q') \cdot Q - (z - \alpha)Q \cdot Q' \\ &= Q^2 + (z - \alpha)Q'Q - (z - \alpha)QQ' = Q^2. \end{aligned} \quad \square$$

2 Recovery of the Turán identity

Corollary 2.1 (Wronskian decomposition of $W(P, P')$).

$$W(P, P') = P \cdot P'' - P'^2 = -\sum_{j=1}^n Q(\alpha_j, z)^2 = -\sum_{j=1}^n W(P, Q_j). \quad (2)$$

Proof. From $P' = \sum Q_j$ (universality) and linearity of the Wronskian in the second argument:

$$W(P, P') = W\left(P, \sum_j Q_j\right) = \sum_j W(P, Q_j) = \sum_j Q_j^2.$$

Since $W(P, P') = PP'' - P'^2$, we get $P'^2 - PP'' = \sum Q_j^2$, which is the Turán–Pandrosion identity (Paper 56). $\square$

Remark 2.2. This provides a *one-line proof* of the Turán identity: the SOS decomposition is simply the linearity of the Wronskian applied to $P' = \sum Q_j$.

3 The cross-Wronskian

Theorem 3.1 (Cross-Wronskian of quotients). *For distinct roots $\alpha_i \neq \alpha_j$:*

$$\boxed{W(Q_i, Q_j) = (\alpha_i - \alpha_j) R_{ij}(z)^2}, \quad (3)$$

where $R_{ij}(z) = P(z)/((z - \alpha_i)(z - \alpha_j))$ is the double Pandrosion quotient.

Proof. Write $Q_i = (z - \alpha_j)R_{ij}$ and $Q_j = (z - \alpha_i)R_{ij}$. Then:

$$\begin{aligned} Q'_i &= R_{ij} + (z - \alpha_j)R'_{ij}, \\ Q'_j &= R_{ij} + (z - \alpha_i)R'_{ij}. \end{aligned}$$

$$\begin{aligned} W(Q_i, Q_j) &= Q'_i Q_j - Q_i Q'_j \\ &= (R_{ij} + (z - \alpha_j)R'_{ij})(z - \alpha_i)R_{ij} - (z - \alpha_j)R_{ij}(R_{ij} + (z - \alpha_i)R'_{ij}) \\ &= (z - \alpha_i)R_{ij}^2 - (z - \alpha_j)R_{ij}^2 \\ &= (\alpha_j - \alpha_i) \cdot (-1) \cdot R_{ij}^2 \cdot (-1) = (\alpha_i - \alpha_j) R_{ij}^2. \quad \square \end{aligned}$$

4 The Pandrosion Wronskian matrix

Definition 4.1. The *Pandrosion Wronskian matrix* is the $n \times n$ antisymmetric matrix:

$$\mathbf{W}(z)_{ij} = W(Q_i, Q_j) = (\alpha_i - \alpha_j) R_{ij}(z)^2. \quad (4)$$

Diagonal entries are zero: $W(Q_i, Q_i) = 0$.

Proposition 4.2. $\mathbf{W}(z)$ factors as:

$$\mathbf{W}(z)_{ij} = (\alpha_i - \alpha_j) \cdot R_{ij}(z)^2 = V_{ij} \cdot G_{ij}(z), \quad (5)$$

where $V_{ij} = \alpha_i - \alpha_j$ is the antisymmetric Vandermonde factor and $G_{ij} = R_{ij}^2$ is the symmetric Gram factor.

Remark 4.3 (Pfaffian conjecture). For even n , the Pfaffian of $\mathbf{W}$ satisfies:

$$\text{Pf } \mathbf{W}(z) = \Delta(\alpha) \cdot \prod_{i < j} R_{ij}(z)^2, \quad (6)$$

where $\Delta(\alpha) = \prod_{i < j} (\alpha_i - \alpha_j)$ is the Vandermonde determinant, and the product of double quotients encodes the z -dependence. At $z = \alpha_k$, most R_{ij} remain nonzero, giving a nontrivial evaluation.

5 Verification

Symbolic ($n = 3, 4, 5$): $W(P, Q_j) = Q_j^2$ verified exactly for all roots.

Decomposition ($n = 3, 4$): $W(P, P') + \sum Q_j^2 = 0$ verified symbolically.

Cross-Wronskian ($n = 4$): $W(Q_i, Q_j) = (\alpha_i - \alpha_j) R_{ij}^2$ verified:

(i, j)	$\alpha_i - \alpha_j$	$W(Q_i, Q_j)$
(1, 2)	-1	$-(z - 3)^2(z - 4)^2$
(1, 3)	-2	$-2(z - 2)^2(z - 4)^2$
(1, 4)	-3	$-3(z - 2)^2(z - 3)^2$
(2, 3)	-1	$-(z - 1)^2(z - 4)^2$
(2, 4)	-2	$-2(z - 1)^2(z - 3)^2$
(3, 4)	-1	$-(z - 1)^2(z - 2)^2$

Numerical: 20,000 random tests, relative error $< 10^{-9}$.

6 Summary

Identity	Formula
Fundamental	$W(P, Q_j) = Q_j^2$
Turán (from decomposition)	$P'^2 - PP'' = \sum Q_j^2 = -W(P, P')$
Cross-Wronskian	$W(Q_i, Q_j) = (\alpha_i - \alpha_j)R_{ij}^2$
Mason–Stothers (Paper 60)	$W(a, b)/ab = F_a - F_b$

References

- [1] E. Borel, *Leçons sur les fonctions entières*, Gauthier-Villars, 1900.
- [2] I. Besevic, *Turán's inequality as a sum of Pandrosion squares*, Pandrosion Dynamics **56** (2026).
- [3] I. Besevic, *Mason–Stothers via the Pandrosion field*, Pandrosion Dynamics **60** (2026).